# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: CSA16  COURSE CODE: Data Warehousing and Data Mining**

**COURSE OUTCOME COVERED**

**CO5:** Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 30%

**QUESTIONS: 05**  **Total Marks:50**

Annexure A
**MOCK INTERVIEW**

| ***Criteria*** | Technical Knowledge ***(2.5)*** | Problem Solving ***(2.5)*** | Analytical Thinking ***(1.5)*** | ***Communication(1.5)*** | Confidence ***(1)*** | Response to Questions(1) | ***Total(10)*** |
|---|---|---|---|---|---|---|---|
| *Weightage* | *25%* | 25% | 15% | 15% | 10% | *10%* | *100%* |
| *Q1* | | | | | | | |
| *Q2* | | | | | | | |

***Code of Conduct:***

*IG Madhulatha(192311295)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***
*G Madhulatha*

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Technical Knowledge | 25 | Demonstrates strong and accurate subject knowledge with in-depth understanding | Shows good knowledge with minor gaps | Basic knowledge with noticeable errors | Lacks fundamental technical knowledge |
| Problem Solving | 25 | Solves problems logically and applies concepts effectively in real scenarios | Applies concepts with moderate accuracy | Limited problem-solving ability; requires guidance | Unable to solve or apply concepts |
| Analytical Thinking | 15 | Analyzes questions critically and provides well-structured, logical answers | Provides reasonable analysis with some structure | Superficial or partially structured responses | No analytical thinking demonstrated |
| Communication | 15 | Communicates clearly, confidently, and professionally with proper terminology | Communicates well with minor hesitation | Communication lacks clarity or confidence | Unable to communicate effectively |
| Confidence | 15 | Displays high confidence, positive attitude, and professional behavior throughout | Shows good confidence with minor issues | Low confidence; inconsistent behavior | Lacks confidence and professionalism |
| Response to Questions | 10 | Responds accurately, promptly, and elaborates with relevant examples | Responds correctly but with limited depth | Partial responses; needs prompting | Unable to respond appropriately |

**Annexure A**

**Mock Interview – Sample Questions (5 Questions)**
(Each question aligned with BL3–BL4, 0.75 Marks each)

**Q1. Dataset: Retail Transactions**

| TID | Items |
|---|---|
| T1 | Milk, Bread |
| T2 | Milk, Butter |
| T3 | Bread, Butter |
| T4 | Milk, Bread, Butter |
| T5 | Milk, Eggs |

Question:
Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example. (BL3 – 0.75 Marks)

**Q2. Dataset: Transaction Data**

| TID | Items |
|---|---|
| T1 | A, B |
| T2 | B, C |
| T3 | A, C |
| T4 | A, B, C |
| T5 | B, C |

Question:
Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used. (BL4 – 0.75 Marks)

**Q3. Dataset: Transaction Data**

| TID | Items |
|---|---|
| T1 | A, B |
| T2 | A |
| T3 | B |
| T4 | A, B |
| T5 | A |

Question:
Explain support and confidence using the dataset and describe their importance in association rule mining. (BL3 – 0.75 Marks)

**Q4 . Dataset: Multilevel Data**

| TID | Items |
|---|---|
| T1 | Electronics, Mobile |
| T2 | Electronics, Laptop |
| T3 | Electronics, Mobile |
| T4 | Electronics, Mobile |
| T5 | Electronics, Laptop |

Question:
Explain multilevel association rules using the dataset and compare them with multidimensional rules. (BL4 – 0.75 Marks)

**Q5. Dataset: Large Transaction Sample**

| TID | Items |
|---|---|
| T1 | A, B, C |
| T2 | A, B |
| T3 | B, C |
| T4 | A, C |
| T5 | A, B, C |

Question:
Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues. (BL4 – 0.75 Marks)

***Q1. Retail Transactions***
*Association rule mining finds relationships between products that customers frequently purchase together.*
*From the given data,* ***Milk and Bread*** *occur together in T1 and T4. Therefore, the store can identify a relationship between these products.*
***Practical example:***
*The store can place* ***Bread near Milk*** *or offer a discount such as* ***"Buy Milk and get Bread at a reduced price."*** *This encourages customers to purchase both products and can increase overall sales.*

***Q2. Transaction Data***
***Difference between Apriori and FP-Growth***
***Apriori:***

- *Generates candidate itemsets level by level.*
- *Requires multiple scans of the database.*
- *Uses the minimum support to eliminate infrequent itemsets.*
- *Simple and easy to understand.*
- *Suitable for **small datasets**.*

***FP-Growth:***

- *Uses an **FP-tree** to store transaction information.*
- *Does not generate large numbers of candidate itemsets.*
- *Requires fewer database scans.*
- *Generally faster than Apriori for large datasets.*
- *Suitable for **large datasets**.*

***Using the given dataset***
*The frequent relationships such as **B and C** can be discovered by both algorithms.*
*For example, B and C occur together in:*
*T2, T4 and T5*
*Support:*
*3/5 × 100 = **60%***
*Therefore, if the minimum support is 60%, **{B, C} is a frequent itemset**.*
***Conclusion***
*Use **Apriori** when the dataset is small and simplicity is important. Use **FP-Growth** when the dataset is large and faster frequent-itemset mining is required.*

Q1. Retail Transactions
Answer:
Association rule mining finds relationships between products that are frequently purchased together.
From the given dataset, Milk and Bread occur together in T1 and T4.

- Milk occurs in T1, T2, T4 and T5.
- Bread occurs in T1, T3 and T4.
- Milk and Bread occur together in T1 and T4.

Therefore, the store can identify a relationship between Milk and Bread.
Practical example:
The store can place Bread near Milk or provide an offer such as "Buy Milk and get Bread at a reduced price."
This encourages customers to purchase both products and can increase overall sales.
Short interview answer:

> Association rule mining identifies products that customers frequently buy together. Here, Milk and Bread occur together, so the store can place them nearby or offer a combo discount to increase sales.

Q2. Transaction Data
Difference between Apriori and FP-Growth

| Apriori | FP-Growth |
|---|---|
| Generates candidate itemsets level by level. | Uses an FP-tree to represent transactions. |
| Requires multiple database scans. | Requires fewer database scans. |
| Generates candidate itemsets. | Does not generate large numbers of candidate itemsets. |
| Simple and easy to understand. | More efficient for large datasets. |
| Suitable for small datasets. | Suitable for large datasets. |

Using the given dataset
The itemset {B, C} occurs together in:

- T2 → B, C
- T4 → A, B, C
- T5 → B, C

Therefore:
Support({B,C}) = 3/5 × 100 = 60%
So, if the minimum support is 60%, {B,C} is a frequent itemset.

Conclusion

Apriori is preferred for small datasets because it is simple and easy to understand. FP-Growth is preferred for large datasets because it is generally faster and avoids generating many candidate itemsets.

Q3. Support and Confidence

Given Dataset

| TID | Items |
|---|---|
| T1 | A, B |
| T2 | A |
| T3 | B |
| T4 | A, B |
| T5 | A |

1. Support

Support tells us how frequently an itemset occurs in the entire transaction database.

For {A,B}:

{A,B} occurs in T1 and T4.

Therefore:

Support(A,B) = 2/5 × 100 = 40%

2. Confidence

Confidence tells us how often B is purchased when A is purchased.

A occurs in:

- T1
- T2
- T4
- T5

So A occurs 4 times.

A and B occur together 2 times.

Therefore:

Confidence(A → B) = Support(A,B) / Support(A)

= 2/4 × 100

= 50%

Importance

- Support identifies frequently occurring itemsets.
- Confidence measures the strength of an association rule.
- They help retailers identify useful product relationships.

Short interview answer:

Support tells how frequently an itemset occurs, while confidence tells how reliably one item is associated with another. In this dataset, support of {A,B} is 40%, and confidence of A → B is 50%.

Q4. Multilevel Association Rules

Given Dataset

| TID | Items |
|---|---|
| T1 | Electronics, Mobile |
| T2 | Electronics, Laptop |
| T3 | Electronics, Mobile |
| T4 | Electronics, Mobile |
| T5 | Electronics, Laptop |

Multilevel Association Rules

Multilevel association rules discover relationships at different levels of abstraction in a concept hierarchy.

For example:

Electronics → Mobile

Here, Electronics is a higher-level category and Mobile is a more specific product category.

From the dataset:

- Electronics occurs in all 5 transactions.
- Mobile occurs in T1, T3 and T4.
- Laptop occurs in T2 and T5.

Therefore, a rule such as:

Electronics → Mobile

can be used to identify relationships between a general product category and a specific product.
Multilevel vs Multidimensional

| Multilevel Association Rules | Multidimensional Association Rules |
|---|---|
| Work at different levels of a concept hierarchy. | Use multiple attributes/dimensions. |
| Example: Electronics → Mobile. | Example: Age = Young → Buys Mobile. |
| Focus on different levels of detail. | Focus on relationships among different attributes. |
| Useful for category-level and product-level analysis. | Useful for customer/profile-based analysis. |

Short interview answer:

> Multilevel association rules discover relationships at different levels of a hierarchy, such as Electronics → Mobile. Multidimensional rules use different attributes such as age, income, product and location.

Q5. Large Transaction Dataset
Given Dataset

| TID | Items |
|---|---|
| T1 | A, B, C |
| T2 | A, B |
| T3 | B, C |
| T4 | A, C |
| T5 | A, B, C |

Challenges in Mining Large Datasets
Mining large datasets can create several problems:

- Large number of transactions
  Processing millions of transactions requires more time.
- Large number of candidate itemsets
  Algorithms such as Apriori can generate many candidate combinations.
- High memory usage
  Storing and processing large datasets requires more memory.
- Multiple database scans
  Repeatedly scanning a large database increases processing time.

How scalable methods solve these problems
FP-Growth can be used because it stores transactions in an FP-tree and avoids generating a large number of candidate itemsets.
Other scalable approaches can use:

- Data partitioning
- Parallel processing
- Distributed computing
- Efficient data structures
- Reduced database scans

Example from the dataset
The itemset {A,B,C} occurs in:

- T1
- T5

Therefore:
Support({A,B,C}) = 2/5 × 100 = 40%
If the minimum support is 40%, {A,B,C} is a frequent itemset.
Conclusion

> Large datasets create problems such as high computation time, memory requirements and many candidate itemsets. Scalable algorithms such as FP-Growth, along with parallel and distributed processing, can reduce these problems and make frequent-itemset mining faster.